Status: **Fully Connected**

System Architecture & Workflow Reference Sheet

Comprehensive architectural blueprint, end-to-end user workflows, recruitment pipeline state machine, and endpoint registry for the WorkWise Editorial Job Board.

Django 5/6 • HTMX • SQLite/Postgres

1. System Architecture & Component Mapping

WorkWise utilizes a modular monolith architecture with specialized domain applications: **accounts** (authentication, dual-role profiles, security), **jobs** (listings, sector classification, preview generation), **applications** (submission snapshots, Kanban pipeline, interview engine), **moderation** (flags, audit logging), and **core** (routing, home, career acceleration resources, persistent notifications, protected media).

Layer	Technologies	Architectural Responsibility & Flow Connection
Presentation	Django Templates, HTMX 1.9+, Vanilla CSS Design Tokens	Dynamic UI without full-page reloads. HTMX powers the real-time job preview card, Kanban column stage transitions, applicant drawer slide-over, and persistent notification polling.
Security & Proxy	Protected Media View, Rate Limiting, RBAC Decorators	All resume downloads route through core.media_views.protected_media_serve. Access is strictly granted to the owning applicant, hiring employer, or superadmin. Rate limiting caps logins at 5 attempts / 5 mins (HTTP 429).
Service Layer	applications.services, accounts.backends, core.notifications	Encapsulates business rules: enforced pipeline state transitions (ALLOWED_TRANSITIONS), interview scheduling, official hire notifications, and dual email/username authentication backend.
Data Layer	SQLite (Dev) / PostgreSQL (Prod), WhiteNoise, Gunicorn	Database models with composite indexes (status, job, applicant), unique application constraints, point-in-time resume snapshotting, and transactional audits.

2. Role-Based Access Control (RBAC) & Permission Matrix

Role / Persona	Status / Level	Primary URL	Authorized Capabilities & System Boundaries
Applicant	`active`	`/applicant/`	Browse sector directories, submit applications with resume snapshot, manage saved profile, track timeline, view interview invitations, download career blueprints.
Employer (Pending)	`pending`	`/employer/pending/`	Newly registered organization awaiting administrative trust verification. Can view pending dashboard but cannot publish live jobs or view applicant submissions.
Employer (Verified)	`active`	`/employer/`	Post/edit/close jobs, preview live cards via HTMX, manage candidate Kanban board, inspect candidate drawer, schedule interviews, log interview outcomes, execute official hires.
Staff Moderator	`moderator`	`/admin-panel/`	Review user content flags (spam, scam, misleading), view reported jobs, monitor system audit trails, view pending employer verification queue.
Superadmin	`superadmin`	`/admin-panel/` `/admin/`	Full platform sovereignty: approve/reject employers, suspend/reactivate users, remove infringing listings, inspect immutable audit logs, direct Django admin access.

3. End-to-End System Workflows

Step-by-step lifecycle tracking how actors interact with each subsystem.

ding	**1. Signup (/accounts/register/)**: User selects role. Applicants receive immedia directed to /employer/pending/. **2. Login (/accounts/login/)**: Dual identifier lookup supports email or username login_attempts_{IP}. Rate limit triggers HTTP 429 after 5 failed attempts within 3 **3. Post-Auth Routing (/accounts/redirect/)**: Evaluates user.role • Employer → /applicant/. **4. Password Reset & Verification**: 4-stage secure flow utilizing 6-digit verificat
IX Preview	**1. Employer Job Creation (/jobs/post/)**: Employer inputs title, department, sec that salary max ≥ salary min. **2. Real-time Preview (/jobs/preview/)**: Form changes trigger an HTMX POST rendering the listing exactly as candidates will see it. **3. Publication & Lifecycle**: Jobs default to draft or open. Employers can close,
Timeline	**1. Discovery (/jobs/)**: Paginated filtering by sector (IT, Healthcare, Education, E (Army, Navy, Coast Guard, Police), remote flag, and salary. **2. Application Submission (/applications/apply/<id>/)**: Validates resume form resume_snapshot. Database constraint blocks duplicate applications. **3. Tracking (/applications/)**: Candidate dashboard renders live timeline chips r
& Official Hire	**1. Pipeline Board (/jobs/<slug>/pipeline/)**: Seven-column Kanban board grou enforce allowed transitions. **2. Slide-Over Candidate Drawer (/applications/<pk>/drawer/)**: Displays cand rating, and internal notes. **3. Interview Scheduling (/applications/<pk>/schedule-interview/)**: Creates I publishes ticket badge on candidate dashboard. **4. Official Hire Confirmation**: Employer specifies confirmed start_date and per and delivers an immediate notification to candidate.
Audit Logging	**1. Verification Queue (/admin-panel/verifications/)**: Admin reviews pending e Reject. **2. Moderation & Content Flags (/moderation/flags/)**: Users flag suspicious lis **3. User Suspension (/admin-panel/users/<pk>/suspend/)**: Superadmin suspe on login. **4. Immutable Audit Trail (/admin-panel/audit-log/)**: All administrative decision target, and timestamp.

4. Recruitment Pipeline State Machine & Transition Rules

Candidate progression is strictly controlled in applications.services.ALLOWED_TRANSITIONS. Any transition outside this specification raises a ValidationError preventing state corruption.

Current Status	Allowed Target Transitions	Business Logic & Automation Triggered
`submitted`	`under_review, shortlisted, rejected`	Initial application state upon submission. Resume snapshot locked.
`under_review`	`shortlisted, interview_scheduled, rejected`	Hiring committee reviewing candidate portfolio and resume.
`shortlisted`	`interview_scheduled, offer_extended, hired, rejected, under_review`	Qualified candidate pool. Employer can revert to review if necessary.
`interview_scheduled`	`offer_extended, hired, shortlisted, rejected`	Active interview slot scheduled. Displays meeting links to applicant.
`offer_extended`	`hired, rejected`	Formal employment proposal presented to candidate.
`hired`	**None (Terminal State)**	Requires start_date & hire_message. Dispatches celebratory in-app notification with onboarding schedule.
`rejected`	under_review *(Re-open candidate)*	Candidate notified. Employer can optionally restore candidate back to review.
`withdrawn`	**None (Terminal State)**	Self-service candidate action. Sets withdrawn_at timestamp.

5. Master URL Routing & Endpoint Directory

Comprehensive mapping of all registered system URLs across applications.

Category	URL Pattern	View Handler	Description & Access Rule
Core	`/`	`core.views.home`	Public landing page with featured jobs, sectors & stats.
Core	`/dashboard/`	`core.views.dashboard`	Authenticated role router redirecting to role dashboard.
Core	`/applicant/`	`core.views.applicant_dashboard`	Applicant hub: application stats, upcoming interviews.
Core	`/employer/`	`core.views.employer_dashboard`	Employer overview: active jobs, applicant counts.
Core	`/employer/pending/`	`core.views.employer_pending`	Verification holding screen for unapproved employers.
Core / Career	`/resume-blueprint/`	`core.views.resume_blueprint`	ATS resume templates and downloadable industry formats.
Core / Career	`/resume-blueprint/interactive/`	`core.views.resume_blueprint_interactive`	Interactive in-browser editable Navy Blueprint resume.
Core / Career	`/interview-mastery/`	`core.views.interview_mastery`	STAR framework guide and interview preparation handbook.
Core / Notif	`/notifications/state/`	`core.notifications.notifications_state`	HTMX endpoint returning unread count and dropdown list.
Accounts	`/accounts/login/`	`accounts.views.LoginView`	Authentication view with rate limiting (5 attempts/5 mins).
Accounts	`/accounts/register/`	`accounts.views.SignupView`	Dual-role registration toggle (Applicant / Employer).
Accounts	`/accounts/profile/`	`accounts.views.profile_view`	Manage avatar, bio, headline, skills, and master resume.
Accounts	`/accounts/password-reset/`	`accounts.views.WorkWisePasswordResetView`	4-stage verification code password reset process.
Jobs	`/jobs/`	`jobs.views.job_list`	Searchable job board with sector, branch, salary filters.
Jobs	`/jobs/manage/`	`jobs.views.employer_job_list`	Employer listing manager (Draft, Open, Closed, Removed).
Jobs	`/jobs/post/`	`jobs.views.job_create`	Post new job with instant HTMX preview card support.
Jobs	`/jobs/preview/`	`jobs.views.job_preview_card`	HTMX live preview card partial generator.
Jobs	`/jobs/<slug>/pipeline/`	`jobs.views.job_pipeline`	Interactive Kanban applicant review board.
Applications	`/applications/apply/<id>/`	`applications.views.apply`	Application submit engine with point-in-time resume snapshot.
Applications	`/applications/`	`applications.views.my_applications`	Applicant submission history with stage chips.
Applications	`/applications/<pk>/drawer/`	`applications.views.application_drawer`	Slide-over candidate inspection panel.
Applications	`/applications/<pk>/transition/`	`applications.views.application_transition`	Enforce status change, official hire message & start date.
Applications	`/applications/<pk>/schedule-interview/`	`applications.views.schedule_interview_view`	Schedule interview slot and emit ticket.
Admin Panel	`/admin-panel/`	`admin_panel_views.admin_panel_dashboard`	Platform operations dashboard and metric tiles.
Admin Panel	`/admin-panel/verifications/`	`admin_panel_views.admin_verification_queue`	Approve or reject pending employer registrations.
Admin Panel	`/admin-panel/users/<pk>/suspend/`	`admin_panel_views.admin_user_suspend`	Superadmin action to suspend offending users.
Admin Panel	`/admin-panel/audit-log/`	`admin_panel_views.admin_audit_log`	Immutable platform audit log of administrator actions.
Security	`/media/<path>`	`core.media_views.protected_media_serve`	Permission-gated file serving proxy for uploaded resumes.

6. Relational Data Models & Key Schema Architecture

Core database entities and foreign-key relationships supporting platform operations.

Model Entity	Application	Key Fields & Schema Attributes	Relational Connections & Constraints
User	`accounts`	role (applicant, employer, admin), status (active, pending, suspended), access_level (moderator, superadmin), avatar, bio, phone, location.	Extends AbstractUser. 1:1 with EmployerProfile or ApplicantProfile. Target of all audit logs and notifications.
EmployerProfile	`accounts`	company_name, company_size, industry, founded_year, linkedin_url, company_logo.	OneToOneField to User (role='employer'). Verification status tied to user status.
ApplicantProfile	`accounts`	headline, resume (master), experience_level, skills (CSV), available_from, portfolio/social links.	OneToOneField to User (role='applicant'). Master resume validated by validate_resume_file.
Job	`jobs`	title, slug, status (draft, open, paused, closed, filled, removed), job_field, uniformed_branch, department, job_type, salary_min, salary_max.	FK to User (employer). FK to Category (null=True). Has many JobTag and Application entities.
Application	`applications`	status, cover_letter, resume_snapshot, rating (1-5), employer_notes, start_date, hire_message, applied_at.	FK to User (applicant), FK to Job. **UniqueConstraint(applicant, job)**. Has many Interview records.
Interview	`applications`	scheduled_at, duration_minutes, format (phone, video, onsite, technical), meeting_link, outcome.	FK to Application. Generates upcoming interview cards on applicant and employer dashboards.
Notification	`core`	kind, message, highlight, detail_title, description, is_read, read_at, pills, action_url.	FK to User. Read state managed server-side. Polled or updated reactively via HTMX.
AuditLog	`moderation`	actor, action (approve, reject, suspend, reactivate, remove), target_type, target_id, ip_address, timestamp.	Immutable compliance log tracking every significant moderation and administrative action.

7. Demonstration Seed Accounts & Execution Reference

Pre-configured testing accounts with specific roles, profiles, and initial test data.

Role	Login Identifier	Password	Entity / Profile Name	Initial State / Target Dashboard
Superadmin	`admin@workwise.dev` *(or 'admin')*	`admin1234`	Platform Superadmin	Redirects to /admin-panel/. Full access to user management, verification, audit logs.
Employer 1	`employer1@workwise.dev` *(or 'employer1')*	`employer1234`	**Synapse Labs** (Metro Health & Tech)	Redirects to /employer/. Verified employer with open job postings & active candidate pipeline.
Employer 2	`employer2@workwise.dev` *(or 'employer2')*	`employer1234`	**Orbit Systems** (National Defense Agency)	Redirects to /employer/. Verified employer with candidate submissions under review.
Applicant 1	`applicant1@workwise.dev` *(or 'applicant1')*	`applicant1234`	Alex Rivera / Chen (Cloud Infrastructure)	Redirects to /applicant/. Has submitted applications across multiple technical roles.
Applicant 2	`applicant2@workwise.dev` *(or 'applicant2')*	`applicant1234`	Sam Chen / Sarah Johnson (Critical Care Nursing)	Redirects to /applicant/. Candidate with active interview schedule ticket.

Developer Quick Commands Reference

- **Run Dev Server:** python manage.py runserver
- **Re-seed Demo Data:** python manage.py seed_demo --flush *(Admin: admin1234 / Others: password123)*
- **Run Test Suite:** python manage.py test • Specific app: python manage.py test applications
- **Production Docker Build:** docker build -t inkboard:latest . && docker run -p 8000:8000 inkboard:latest